Vitamin B-12 supplementation improves arterial function in vegetarians with subnormal vitamin B-12 status.

OBJECTIVE: Vegetarians are more vascular-healthy but those with subnormal vitamin B-12 status have impaired arterial endothelial function and increased intima-media thickness. We aimed to study the impact of vitamin B-12 supplementation on these markers, in the vegetarians. DESIGN: Double-blind, placebo controlled, randomised crossover study. SETTING: Community dwelling vegetarians. PARTICIPANTS: Fifty healthy vegetarians (vegetarian diet for at least 6 years) were recruited. INTERVENTION: Vitamin B-12 (500 µg/day) or identical placebo were given for 12 weeks with 10 weeks of placebo-washout before crossover (n=43), and then open label vitamin B-12 for additional 24 weeks (n=41). MEASUREMENT: Flow-mediated dilation of brachial artery (FMD) and intima-media thickness (IMT) of carotid artery were measured by ultrasound. RESULTS: The mean age of the subjects was 45±9 years and 22 (44%) were male. Thirty-five subjects (70%) had serum B-12 levels <150 pmol/l. Vitamin B-12 supplementation significantly increased serum vitamin B-12 levels ($p<0.0001$) and lowered plasma homocysteine ($p<0.05$). After vitamin B-12 supplementation but not placebo, significant improvement of brachial FMD (6.3±1.8% to 6.9±1.9%; $p<0.0001$) and in carotid IMT (0.69±0.09 mm to 0.67±0.09 mm, $p<0.05$) were found, with further improvement in FMD (to 7.4±1.7%; $p<0.0001$) and IMT (to 0.65±0.09 mm; $p<0.001$) after 24 weeks open label vitamin B-12. There were no significant changes in blood pressures or lipid profiles. On multivariate analysis, changes in B-12 ($\beta=0.25$; $p=0.02$) but not homocysteine were related to changes in FMD, ($R=0.32$; F value=3.19; $p=0.028$). CONCLUSIONS: Vitamin B-12 supplementation improved arterial function in vegetarians with subnormal vitamin B-12 levels, proposing a novel strategy for atherosclerosis prevention.